Unknown

pair tunneling

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I. INTRODUCTION

$$\hat{H} = -t \sum_{\langle i,j\rangle,\sigma} (\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}) + U \sum_{i} \hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \tag{1}$$

$$\hat{H} = -t \sum_{\langle i,j\rangle} \hat{a}^{\dagger}_{i}\hat{a}_{j} + \frac{U}{2} \sum_{i} \hat{n}_{i}(\hat{n}_{i} - 1) \tag{2}$$

$$H = \sum_{i=1}^{N} \left[-\frac{1}{2}\frac{\partial^2}{\partial x_i^2} + V(x_i) \right] + \sum_{i<j}^{N} g\delta(x_i - x_j) \tag{3}$$

$$H = \sum_{i=1}^{N} \left[-\frac{1}{2}\frac{\partial^2}{\partial x_i^2} + V(x_i) + g \sum_{j=i+1}^{N} \delta(x_i - x_j) \right] \tag{4}$$

$$\begin{aligned} \hat{\mathcal{H}} = & \sum_{\sigma}^{\{L,R\}} \sum_{i} E_i \hat{a}^{\dagger}_{\sigma i}\hat{a}_{\sigma i} - \sum_{i} J_i(\hat{a}^{\dagger}_{Li}\hat{a}_{Ri} + \hat{a}^{\dagger}_{Ri}\hat{a}_{Li}) \\ & + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} U_{\alpha\beta\gamma\delta}\hat{a}^{\dagger}_{\alpha}\hat{a}^{\dagger}_{\beta}\hat{a}_{\gamma}\hat{a}_{\delta} \end{aligned} \tag{5}$$

$$\begin{aligned} \hat{\mathcal{H}} = & \sum_{\sigma}^{\{L,R\}} \sum_{i} E_i \hat{a}^{\dagger}_{\sigma i}\hat{a}_{\sigma i} - \sum_{i} J_i(\hat{a}^{\dagger}_{Li}\hat{a}_{Ri} + \hat{a}^{\dagger}_{Ri}\hat{a}_{Li}) \\ & + \frac{1}{2} \sum_{IJKL} U_{IJKL}\hat{a}^{\dagger}_{I}\hat{a}^{\dagger}_{J}\hat{a}_{K}\hat{a}_{L} \end{aligned} \tag{6}$$

$$\begin{aligned} \hat{\mathcal{H}} = & \sum_{\sigma}^{\{L,R\}} \sum_{i} E_i \hat{a}^{\dagger}_{\sigma i}\hat{a}_{\sigma i} - \sum_{i} J_i(\hat{a}^{\dagger}_{Li}\hat{a}_{Ri} + \hat{a}^{\dagger}_{Ri}\hat{a}_{Li}) \\ & + \frac{1}{2} \sum_{KLMN} U_{KLMN}\hat{a}^{\dagger}_{K}\hat{a}^{\dagger}_{L}\hat{a}_{M}\hat{a}_{N} \end{aligned} \tag{7}$$

$$\begin{aligned} \hat{\mathcal{H}}_{\texttt{2Mode}} = & -J(\hat{a}^{\dagger}_{L0}\hat{a}_{R0} + \hat{a}^{\dagger}_{R0}\hat{a}_{L0}) + \frac{U}{2}(\hat{a}^{\dagger 2}_{L0}\hat{a}^{2}_{L0} + \hat{a}^{\dagger 2}_{R0}\hat{a}^{2}_{R0}) + V\hat{a}^{\dagger}_{L0}\hat{a}_{L0}\hat{a}^{\dagger}_{R0}\hat{a}_{R0} \\ & + \sum_{\sigma}^{\{L,R\}} T(\hat{a}^{\dagger}_{L0}\hat{a}^{\dagger}_{\sigma 0}\hat{a}_{\sigma 0}\hat{a}_{R0} + \hat{a}^{\dagger}_{R0}\hat{a}^{\dagger}_{\sigma 0}\hat{a}_{\sigma 0}\hat{a}_{L0}) + \frac{V}{4}(\hat{a}^{\dagger 2}_{L0}\hat{a}^{2}_{R0} + \hat{a}^{\dagger 2}_{R0}\hat{a}^{2}_{L0}) \end{aligned} \tag{8}$$

$$\rho^{(\mathtt{k})}(x,x') = \frac{1}{N}\langle \mathtt{k}|\hat{\Psi}^{\dagger}(x)\hat{\Psi}(x')|\mathtt{k}\rangle \tag{9}$$

$$\rho^{(\mathtt{k})}(x,x') = \sum_i \lambda_i \psi_i^*(x)\psi_i(x') \tag{10}$$

$$\rho^{(|\mathtt{k}\rangle)}(x,x') = \frac{1}{N}\langle \mathtt{k}|\hat{\Psi}^{\dagger}(x)\hat{\Psi}(x')|\mathtt{k}\rangle \tag{11}$$

$$\rho^{(|\mathtt{k}\rangle)}(x,x') = \sum_i \lambda_i \psi_i^*(x)\psi_i(x') \tag{12}$$

$$\rho^{(\mathtt{N})}(x,x') \approx \psi_0^*(x)\psi_0(x') \tag{13}$$

$$\rho^{(|\mathtt{N}\rangle)}(x,x') \approx \psi_0^*(x)\psi_0(x') \tag{14}$$

$$\phi_A(x) = \psi_0(x) \tag{15}$$

$$\phi_A(x) = \psi_0(x) = \sum_i [\lambda_{Li}\varphi_{Li}(x) + \lambda_{Ri}\varphi_{Ri}(x)] \tag{16}$$

$$\phi_A(x) = \sum_i [c_{Li}\varphi_{Li}(x) + c_{Ri}\varphi_{Ri}(x)] \tag{17}$$

$$\phi_A(x) = \sum_i [\lambda_{Li}\varphi_{Li}(x) + \lambda_{Ri}\varphi_{Ri}(x)] \tag{18}$$

$$\phi_B(x) = \sum_i (-1)^i [\langle Ri|\phi_A\rangle\varphi_{Li}(x) - \langle Li|\phi_A\rangle\varphi_{Ri}(x)] \tag{19}$$

$$\phi_B(x) = \sum_i (-1)^i [c_{Ri}\varphi_{Li}(x) - c_{Li}\varphi_{Ri}(x)] \tag{20}$$

$$\phi_B(x) = \sum_i (-1)^i [\lambda_{Ri}\varphi_{Li}(x) - \lambda_{Li}\varphi_{Ri}(x)] \tag{21}$$

$$\begin{aligned}\hat{\mathcal{H}}_{\mathtt{Eff}} = \hat{\mathcal{H}}_{\mathtt{2Mode}} &+ \frac{W}{6}\sum_{\sigma}^{\{L,R\}} [\hat{n}_{\sigma 0}(\hat{n}_{\sigma 0}-1)(\hat{n}_{\sigma 0}-2)] \\ &+ \frac{Q}{2}[(\hat{a}_{L0}^{\dagger 3}\hat{a}_{L0}^2\hat{a}_{R0} + \hat{a}_{R0}^{\dagger 3}\hat{a}_{R0}^2\hat{a}_{L0}) + h.c.]\end{aligned} \tag{22}$$

$$\hat{\mathcal{H}}_{\texttt{Eff}} = \hat{\mathcal{H}}_{\texttt{2Mode}} + \frac{W}{6}[\hat{n}_{L0}(\hat{n}_{L0}-1)(\hat{n}_{L0}-2) + \hat{n}_{R0}(\hat{n}_{R0}-1)(\hat{n}_{R0}-2)] + \frac{Q}{2}[(\hat{a}^{\dagger 3}_{L0}\hat{a}^{2}_{L0}\hat{a}_{R0} + \hat{a}^{\dagger 3}_{R0}\hat{a}^{2}_{R0}\hat{a}_{L0}) + h.c.] \tag{23}$$

$$\hat{\mathcal{H}}_{\texttt{Eff}} = \hat{\mathcal{H}}_{\texttt{2Mode}} + \frac{W}{6}[\hat{n}_{L0}(\hat{n}_{L0}-1)(\hat{n}_{L0}-2) + \hat{n}_{R0}(\hat{n}_{R0}-1)(\hat{n}_{R0}-2)] + \frac{Q}{2}[(\hat{a}^{\dagger}_{L0}\hat{a}^{\dagger}_{L0}\hat{a}^{\dagger}_{L0}\hat{a}_{L0}\hat{a}_{L0}\hat{a}_{R0} + \hat{a}^{\dagger}_{R0}\hat{a}^{\dagger}_{R0}\hat{a}^{\dagger}_{R0}\hat{a}_{R0}\hat{a}_{R0}\hat{a}_{L0}) + h.c.] \tag{24}$$

$$\hat{\mathcal{H}}_{\texttt{Eff}} = \hat{\mathcal{H}}_{\texttt{2Mode}} + \frac{W}{6}\sum_{\sigma}^{\{L,R\}}[\hat{n}_{\sigma}(\hat{n}_{\sigma}-1)(\hat{n}_{\sigma}-2)] + \frac{Q}{2}[(\hat{a}^{\dagger 3}_{L}\hat{a}^{2}_{L}\hat{a}_{R} + \hat{a}^{\dagger 3}_{R}\hat{a}^{2}_{R}\hat{a}_{L}) + h.c.] \tag{25}$$

$$\hat{\mathcal{H}}_{\texttt{Eff}} = \hat{\mathcal{H}}_{\texttt{2Mode}} + \frac{W}{6}[\hat{n}_{L}(\hat{n}_{L}-1)(\hat{n}_{L}-2) + \hat{n}_{R}(\hat{n}_{R}-1)(\hat{n}_{R}-2)] + \frac{Q}{2}[(\hat{a}^{\dagger 3}_{L}\hat{a}^{2}_{L}\hat{a}_{R} + \hat{a}^{\dagger 3}_{R}\hat{a}^{2}_{R}\hat{a}_{L}) + h.c.] \tag{26}$$

$$\hat{\mathcal{H}}_{\texttt{Eff}} = \hat{\mathcal{H}}_{\texttt{2Mode}} + \frac{W}{6}[\hat{n}_{L}(\hat{n}_{L}-1)(\hat{n}_{L}-2) + \hat{n}_{R}(\hat{n}_{R}-1)(\hat{n}_{R}-2)] + \frac{Q}{2}[(\hat{a}^{\dagger}_{L}\hat{a}^{\dagger}_{L}\hat{a}^{\dagger}_{L}\hat{a}_{L}\hat{a}_{L}\hat{a}_{R} + \hat{a}^{\dagger}_{R}\hat{a}^{\dagger}_{R}\hat{a}^{\dagger}_{R}\hat{a}_{R}\hat{a}_{R}\hat{a}_{L}) + h.c.] \tag{27}$$